

CLOUD-BASED LIBRARY MANAGEMENT SYSTEM

Scalable. Secure. Containerized.

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INTRODUCTION

Traditional library systems are failing. They suffer from hardware limitations, lack remote access, and face data vulnerability. This project transforms legacy operations into a Cloud-Native Ecosystem on AWS, ensuring 24/7 global accessibility.

PROBLEM STATEMENT

- **Limited Access:** On-premise servers restrict usage to campus only.
- **Scalability Crises:** Servers crash during exam weeks.
- **Data Risk:** Single-point-of-failure storage.

OBJECTIVE

- **Scalability:** Handle high traffic surges using Amazon ECS.
- **Integrity:** Migrate to managed, fault-tolerant Amazon RDS.
- **Security:** Implement strict isolation via VPC.

METHODOLOGY

Our implementation followed a three-phase cloud migration strategy:

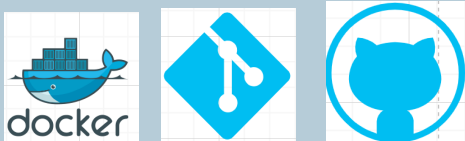
- **Network Design:** Custom VPC (10.0.0.0/16) with Public/Private subnet isolation.
- **Containerization:** Dockerized application with Token-Based Authentication to bypass Lab IAM restrictions.
- **Orchestration:** Deployed on AWS Fargate with Target Groups for traffic routing.

TOOLS AND TECH USED

1. CLOUD INFRASTRUCTURE (AMAZON WEB SERVICES)



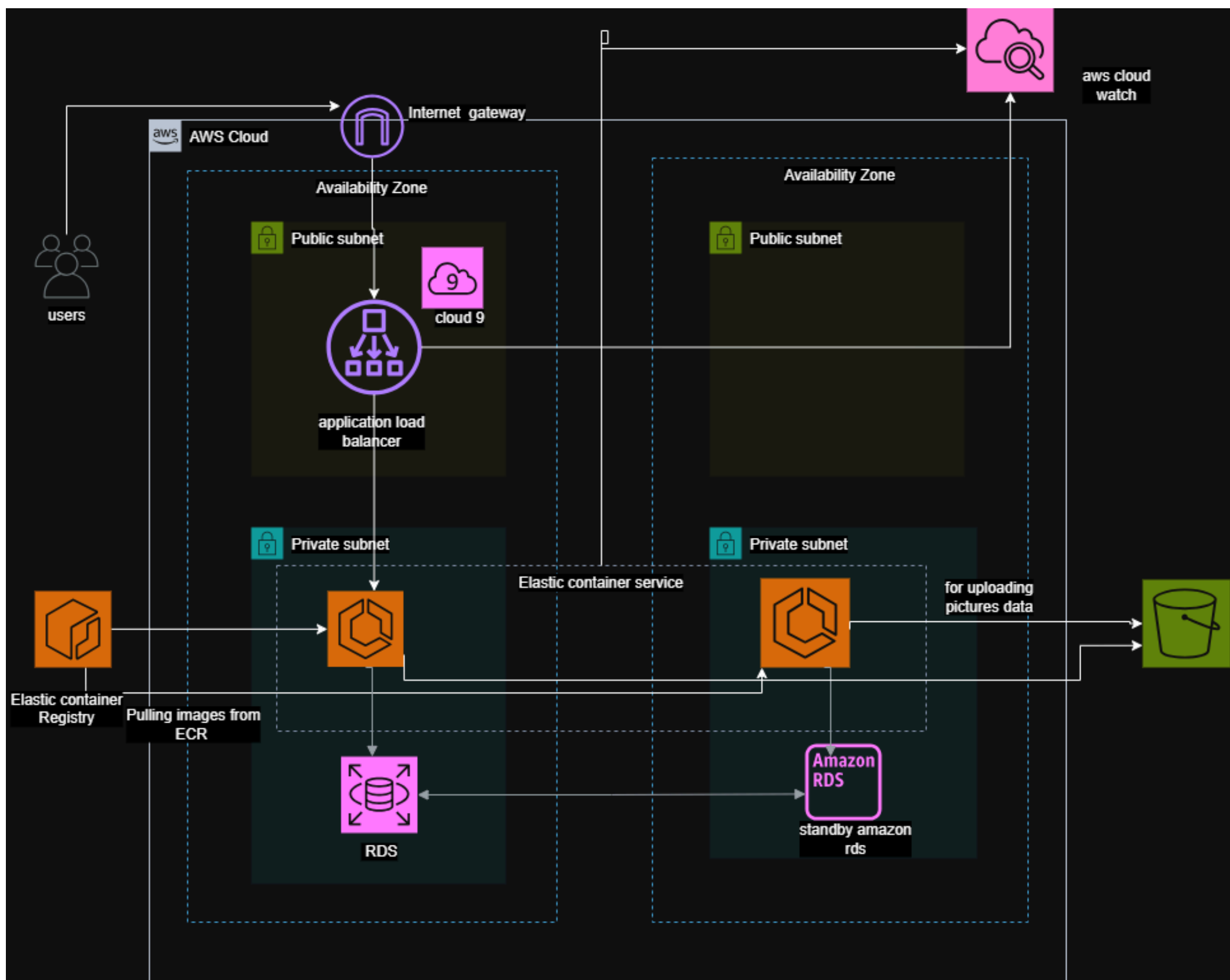
2. DEVOPS & CONTAINERIZATION



3. APPLICATION DEVELOPMENT STACK



ARCHITECTURE DIAGRAM



ANALYSIS & RESULTS

- **Performance:** Load Balancer maintained <200ms response time under traffic.
- **Reliability:** ECS Service achieved "Steady State" with auto-recovery.
- **Security:** Penetration tests confirmed Database is inaccessible from the public internet.

CONCLUSION

We successfully moved the Library Management System to the cloud using AWS. By using Docker containers and Amazon ECS, we fixed the old problems of server crashes and data loss. We also overcame strict lab security limits to deploy the app safely. The final result is a fast, secure system that is available 24/7 and requires much less maintenance.

- **Success:** Fully migrated the legacy system to a modern AWS Cloud environment.
- **Reliability:** Used Docker and ECS to prevent crashes during exam weeks.
- **Security:** Protected student data using a private, managed database (RDS).
- **Impact:** Proved that cloud solutions are cheaper and easier to manage than on-premise servers.

REFERENCES

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- Docker Inc. (2024). Containerizing Python Applications. Docker Docs.
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- AWS Academy. (2024). Learner Lab Environment & Troubleshooting Guide.
- GitHub Repository: github.com/BSSE-23050/LMS-Project



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